Muhammad Ashir

Ilmenau, Germany | +49 1512 3876493 | ashirali0998@gmail.com | LinkeDin | GitHub | Nationality: Pakistani

EDUCATION

Technische Universität Ilmenau, Germany

Master of Research (M.Res.) Computer and Systems Engineering — *Oct 2025 – Present*

Focus Areas: *Artificial Intelligence, Machine Learning, Data Engineering, Big Data Analytics, Distributed Systems, Cloud Computing*

University of Central Punjab, Pakistan

Oct 2020 – Aug 2024

Bachelor of Science in Software Engineering, Relevant

Coursework: *Data Structures & Algorithms, Database Systems, Software Engineering, Object-Oriented Programming, Statistics, Artificial Intelligence*

EXPERIENCE

Software Engineer

Jul 2024 – Sep 2025

Bootlogix

Technologies: Oracle Database, SQL, Apache Airflow, ETL Pipelines, Jenkins, Banking Systems

- Designed and managed end-to-end ETL pipelines for banking systems involving more than 500 source tables.
- Developed workflow orchestration solutions using Apache Airflow, reducing manual processing effort by 60%.
- Optimized Oracle databases containing over 500 million records through indexing, partitioning, and query tuning.
- Improved ETL performance and throughput by 40% while maintaining strict regulatory compliance requirements.
- Participated in requirements gathering, testing, deployment, debugging, and production support.
- Collaborated with cross-functional teams to deliver reliable financial data solutions.

Software Engineer

Sep 2023 – May 2024

CubeBytes Solutions

Client: Central Bank of Egypt | Technologies: Oracle Database, Oracle Analytics Server, SQL, PL/SQL, KYC Systems

- Developed Oracle-based data solutions supporting financial transaction processing systems.
- Built and maintained data pipelines for regulatory reporting and analytics.
- Designed and deployed 20+ dashboards using Oracle Analytics Server.
- Implemented KYC validation mechanisms that reduced manual compliance effort by 25%.
- Improved onboarding efficiency and reporting accuracy through automation and system integration.
- Worked closely with stakeholders to deliver enterprise reporting solutions.

PROJECTS

Access Bank Ghana – Core Banking | Associated with Bootlogix | Technologies: SQL, ETL, Data Integration

- Installed, configured, and supported Oracle applications, including troubleshooting and production issue resolution.
- Customized application workflows, views, and components based on business requirements.
- Optimized SQL queries and database workflows to improve application performance.
- Developed ETL scripts for data extraction, transformation, and integration.
- Designed operational reports to support business and system requirements.
- Performed system testing, debugging, and deployment support across project phases.
- Created technical documentation and delivered knowledge-transfer sessions to client teams.
- Collaborated with client stakeholders to support implementation, handover, and project delivery.

NLPowereDub – AI-Based English-to-Urdu Dubbing System | Final Year Project | Technologies: Python, NLP, Machine Learning

- Developed an AI-assisted system for dubbing English videos into Urdu.
- Integrated natural language processing and machine learning models for speech processing and translation.
- Focused on reducing synchronization differences between original and dubbed content.
- Worked on voice alignment and lip reanimation techniques to improve dubbing quality.
- Conducted experimentation and evaluation to improve realism and user experience.

CERTIFICATIONS

Building with the Claude API | 2026 View Certificate

Claude 101 | 2026 View Certificate

AI Fluency: Framework & Foundations | 2026 View Certificate

Introduction to Agent Skills | 2026 View Certificate

OPEN SOURCE & RESEARCH CONTRIBUTIONS

GitHub Research Contributions

- Contributed to research-oriented open-source projects through GitHub.
- Participated in bug fixing, feature improvements, documentation, and testing activities.
- Collaborated using Git workflows, pull requests, code reviews, and issue-tracking systems.

RESEARCH INTERESTS

- Artificial Intelligence - Machine Learning - Data Engineering - Cloud Computing - Financial Technology - LLM

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScriptython | **Data:** ETL/ELT, Airflow, Data Modeling, Data Integration, Data Quality, Workflow Automation | **Databases:** Oracle, PostgreSQL, MySQL | **AI/ML:** TensorFlow, PyTorch, Machine Learning, NLP, Data Preprocessing, Model Evaluation | **Cloud/DevOps:** AWS, Docker, Git, GitHub Actions, Linux

LANGUAGES

English — C1 | German — A2 (actively progressing toward B1) | Urdu — Native

ADDITIONAL INFORMATION

Work Authorization: Eligible to work in Germany under student employment regulations.

Availability: Immediately – Werkstudent, HiWi, Internship, Software/Data Engineering & AI roles